Supplementary Material

S1. Purpose and archived provenance

This supplement provides additional methodological, diagnostic, and reproducibility details for the analyses reported in the main manuscript. The natural-shift experiments correspond to software commit `d745e5412c1d530a2ae64e2eaa42c85c1f64e419`; the archived experimental outputs and analysis code correspond to commits `74f56923907b75cac638b411b5af6293e4f3a11e` and `9836d7239f7c96d50e6f510d12fc1e8344752b9c`. All tables are rounded displays of archived machine-readable records, whose source files are listed in `manuscript/source_data_v2/`. Subsequent reproduction checks were conducted separately and did not replace or modify the archived outputs used in the manuscript.

The archived execution contained 54 completed model configurations and no technical failure row. The six tasks, five source families, two base-model types, three split seeds, and frozen model-seed strata are described in the main paper. Positive-budget target samples used 100 adaptation seeds per configuration. The analyses reported in the manuscript use the archived outputs. A subsequent representative ACS reproduction check was run separately and did not replace or modify those outputs. An earlier probability-simulation replay added missing calibration metrics before the archived analysis version and was not repeated for manuscript preparation.

S2. Entity, domain, and role controls

The entity audit reported equality of row count and unique-entity count for every final task table, with zero duplicated entities. The entity units were person, household, respondent, respondent, patient, and institution for ACS Income, ACS Food Stamps, BRFSS, NHANES, UCI diabetes, and College Scorecard, respectively. UCI encounters were reduced to a deterministic patient index before role assignment. Role hashes were generated after entity-level assignment, and no role cap used labels.

Seven roles were mutually exclusive for each split: source training, source tuning, source probability calibration, source conformal calibration, source in-domain testing, target adaptation, and target final testing. The separate source calibration roles prevent probability fitting from reusing the scores that define source conformal thresholds. Target final testing was never used for model selection, acquisition, calibration, or threshold fitting. For a given adaptation seed, positive budgets were nested prefixes of one label-blind deterministic ordering, and all methods shared the resulting sample hash.

S2.1 Exact task definitions and source versions

ACS Income used 2018 ACS 1-year person PUMS records and the Folktables adult filter `AGEP > 16`, `PINCP > 100`, `WKHP > 0`, and `PWGTP` $\geq$ `1`; the project label was `PINCP` $\geq$ `56,000`. `DIVISION = 1` (New England) was target and other divisions were source (U.S. Census Bureau 2018, 2019; Ding et al. 2021). ACS Food Stamps joined the person and housing PUMS tables by household serial, then retained `RELP = 0`, `18` $\leq$ `AGEP < 62`, `PINCP` $\leq$ `30,000`, and `HUPAC` $\in$ `{1,2,3,4}`. `FS = 1` was positive; `DIVISION = 6` (East South Central) was target. `PINCP` is personal income. Census `HUPAC` code 4 means no children, so this cohort is not restricted to households with children.

BRFSS used the 2015, 2017, 2019, and 2021 public-use files. `DIABETE3` or `DIABETE4` code 1 was positive; valid codes 2, 3, and 4 were negative, while 7, 9, and missing values were excluded. `_PRACE1 = 1` was source and codes 2–6 were target; this is a race-code partition and not a White-non-Hispanic definition (Centers for Disease Control and Prevention 2015, 2017, 2019, 2021). NHANES merged cycle-qualified `SEQN` identifiers across 1999–2000 through 2017–2018, retained complete `LBXBPB`, `INDFMPIR`, and `RIDAGEYR` at age 18 or younger, labeled `LBXBPB` $\geq$ `3.5` μg/dL, and split target at `INDFMPIR` $\leq$ `1.3` (National Center for Health Statistics 2020a, 2020b, 2020c). The cutpoint is a benchmark informed by the CDC blood lead reference value, not a toxicity or diagnosis threshold, and no survey weights were used (Ruckart et al. 2021).

The UCI task used the public 1999–2008 diabetes encounters, with a stable-hash patient index encounter rather than a chronology claim. `admission_source_id = 7` (emergency room) was target, and `readmitted != NO` was positive, including both `<30` and `>30` categories (Clore et al. 2014; Strack et al. 2014). College used `MERGED2018_19_PP.csv`; `CONTROL = 1` or 2 was source, `CONTROL = 3` was target, and `C150_4` $\leq$ `0.50` was positive. `C150_4` is the applicable IPEDS first-time, full-time cohort completion measure within 150% of normal time, not an all-student pooled completion rate (U.S. Department of Education 2026b, 2026a, 2025).

S3. H1 rate-level feasibility audit

The H1 validation concerns event rates, not the discrepancy between a single binary realization and its probability. The frozen run-level values called MAE and RMSE equal functions of $|Y - p|$ for a single support event and remain nonzero under a correctly specified Bernoulli probability. They are therefore not used here as probability-versus-rate calibration measures. The corrected analysis groups the 100 adaptation draws within each design stratum, compares empirical event rate with its finite-population probability, and then applies task- and family-equal summaries. Model-stratified rows that share one label sample are design duplicates rather than independent repetitions.

Across 1,260 model-stratified groups, the task-equal absolute class-support rate gap was 0.002433, the family-equal gap 0.002658, and the task-equal signed gap +0.000106. Wilson intervals covered the predicted rate in 100% of groups. Algorithm eligibility equaled the method-specific class-support indicator in all 126,000 method–model-stratified available records after score-source deduplication: the observed algorithm-minus-support difference had minimum and maximum zero. This is a technical validation of the archived implementation and hypergeometric support calculation, not a separate theory result. Classwise finite-threshold requirements and sparse-class limitations are established in conformal literature (Angelopoulos and Bates 2023; Ding et al. 2023; Liu et al. 2026; Althani 2026).

Supplementary Table S1. Task-equal empirical support and finite-threshold rates. Predicted finite-threshold rates use the observed target-pool composition retrospectively.

Budget	≥1/class empirical	≥5/class empirical	Mondrian finite empirical	Mondrian finite predicted
25	0.9806	0.6917	0.3306	0.3289
50	0.9983	0.8939	0.7317	0.7392
100	1.0000	0.9856	0.9100	0.9015
250	1.0000	1.0000	0.9993	0.9997
500	1.0000	1.0000	1.0000	1.0000

Across all H1 strata, predicted and empirical finite-threshold means were 0.779157 and 0.779643, the mean absolute split–model-stratum rate gap was 0.010973, and the interval-coverage rate was 97.6%. The larger gap at budget 25 reflects the higher sampling variation of a classwise event. Prospective planning cannot substitute the true target composition used in this retrospective validation. The frozen planning analysis instead uses the prespecified prevalence range [0.01, 0.99], which is deliberately wider than the observed task prevalences and should not be presented as a point prediction for the six tasks.

S4. H2 model-stratified effects

The main H2 table reports an equal mean of the logistic-regression and CPU-XGBoost task-equal rows. Supplementary Table S2 shows the endpoint rows separately. Directions agree closely for sigmoid and isotonic, while the budget-25 intercept means straddle zero. This model separation supports the aggregate ordering without treating the two models as identical. The complete 40-row positive-budget table, including every intermediate budget, is supplied as `source_data_v2/supp_table_s2_h2_model_strata.csv`.

Supplementary Table S2. Model-stratified primary-set endpoint effects.

Model	Method	Budget	Task-equal Δ log loss	Material harm	Estimable
Logistic regression	Intercept MLE	25	-0.000614	44.2%	97.7%
Logistic regression	Intercept MLE	500	$+0.008617$	1.3%	100.0%
CPU XGBoost	Intercept MLE	25	$+0.000409$	42.0%	97.7%
CPU XGBoost	Intercept MLE	500	$+0.008989$	0.8%	100.0%
Logistic regression	Jeffreys intercept	25	-0.000139	44.9%	100.0%
Logistic regression	Jeffreys intercept	500	$+0.008497$	1.3%	100.0%
CPU XGBoost	Jeffreys intercept	25	$+0.000823$	42.2%	100.0%
CPU XGBoost	Jeffreys intercept	500	$+0.008864$	0.7%	100.0%
Logistic regression	Sigmoid	25	-0.026387	74.9%	97.7%
Logistic regression	Sigmoid	500	$+0.007888$	1.8%	100.0%
CPU XGBoost	Sigmoid	25	-0.024515	73.0%	97.7%
CPU XGBoost	Sigmoid	500	$+0.009085$	1.6%	100.0%
Logistic regression	Isotonic	25	-0.439859	99.8%	64.5%
Logistic regression	Isotonic	500	-0.017208	73.3%	100.0%
CPU XGBoost	Isotonic	25	-0.403667	99.7%	64.5%
CPU XGBoost	Isotonic	500	-0.017238	73.0%	100.0%

Harm rates in Table S2 are conditional on successful finite paired effects. Their denominators therefore differ from availability denominators when class support fails. Isotonic at budget 25 is the clearest example: approximately 64.5% of attempts were estimable, and almost all successful pairs were materially harmful. In contrast, Jeffreys smoothing was available in all attempts, but its endpoint performance stayed close to intercept-only MLE. Neither result supports replacing the separate availability and conditional-performance columns with a single score.

S5. H3 budget fractions, deletion checks, and College stress case

At absolute budget 100, the frozen H3 fraction is calculated from the actual split-level target adaptation pool, not by dividing by a conservative registry capacity. The fractions span more than an order of magnitude among the primary five-task set and approach the entire College pool. These values show why an absolute count and a relative fraction should both be reported while neither is treated as a complete task-normalized sample-size measure.

Supplementary Table S3. Budget 100 as a fraction of the executed target adaptation pool.

Task	Analysis scope	Pool fraction	Percent
BRFSS Diabetes	primary set	0.000872	0.087%
ACS Income	primary set	0.002958	0.296%
Diabetes Readmission	primary set	0.006629	0.663%
ACS Food Stamps	primary set	0.011357	1.136%
NHANES Lead	primary set	0.019697	1.970%
College Scorecard	separate stress	0.961538	96.154%

The frozen deletion audit retained 169/170 key LOTO directions and 136/136 LOFO directions. The only reversal was approximately −0.000028, about 72-fold smaller than the prespecified 0.002 material-benefit threshold. These checks recompute the same frozen paired evidence after removing one task or family; they do not create independent shifted populations. Complete unrounded tables are included as `source_data_v2/supp_full_loto.csv` and `source_data_v2/supp_full_lofo.csv`.

Supplementary Table S4. Direction retention under deletion checks.

Check	Retained	Total	Rate
Leave one task out	169	170	0.9941
Leave one source family out	136	136	1.0000

College Scorecard supplies a boundary rather than an additional common-task replication. The executed curve contains only budgets 0, 25, 50, and 100, while 250 and 500 remain unavailable. At the same method and budget, logistic regression and XGBoost could have opposing effect directions. Pooling this task into primary-set averages would therefore mix an almost exhaustive small-pool sample with much smaller sampling fractions and would create incomplete upper-budget curves. The complete 240-row College derived table is supplied as `source_data_v2/supp_college_stress.csv`.

S6. H4 strict-threshold diagnostic

H4 uses raw target-final minus source-ID differences. A positive Brier or log-loss difference means deterioration, whereas a positive AUROC difference means improved discrimination. The strict thresholds exclude equality. All 17 pairs within 0.01 and all 32 pairs within 0.02 worsened on both proper scores. Other diagnostics were less uniform: at the 0.02 threshold, AUPRC decreased in 3/32, absolute calibration-intercept error worsened in 28/32, and slope error worsened in 11/32.

Supplementary Table S5. H4 strict-AUROC-threshold subsets.

Strict	Pairs	Brier worse	Log loss worse	Median Δ Brier	Median Δ log loss	AUPRC down	Slope error worse
$\lvert\Delta\text{AUROC}\rvert < 0.01$	17	17	17	+0.027073	+0.071711	0	5
$\lvert\Delta\text{AUROC}\rvert < 0.02$	32	32	32	+0.018232	+0.046746	3	11

Across all 54 primary pairs, Pearson correlations between AUROC change and Brier/log-loss change were −0.907897/−0.895421 and Spearman correlations −0.761997/−0.772365. Because multiple design pairs arose from the same tasks, these are descriptive associations rather than population-level correlation estimates. The supported diagnostic is narrower: a small AUROC change did not rule out proper-score deterioration. This separation between ranking and calibration is already established in shifted clinical and uncertainty-evaluation work (Davis et al. 2017; Levy et al. 2022; Guo et al. 2023; Ovadia et al. 2019; Van Calster et al. 2019).

S7. H5 low-budget thresholds and doubletons

The primary H5 analysis compares standard and Mondrian conformal prediction on identical task, split, model, model seed, adaptation seed, budget, score source, evaluation split, and sample hash. Metric differences are defined only for jointly successful pairs and are summarized by the median within each task–model–budget cell before equal averaging across cells. At budget 25, the primary joint-success conditional mean difference in worst-class coverage was +0.1285 and the mean set-size difference was +0.3280; this summary retains successful Mondrian outputs with infinite thresholds. Across available attempts, the Mondrian finite-threshold rate was only 30.1%. Restricting additionally to jointly successful pairs with finite Mondrian thresholds reduced the mean differences to about +0.0275 for worst-class coverage and +0.0013 for set size. That sensitivity subset contained four tasks and eight task–model cells; NHANES contributed no finite row at budget 25. Infinite thresholds and full two-label sets therefore account for much of the larger primary conditional difference. Non-estimable attempts were not assigned zero metric differences.

Supplementary Table S6. Budget-25 H5 sensitivity to finite thresholds.

Subset	Δ worst-class coverage	Δ mean set size	Finite-threshold rate in subset
Joint-success pairs (infinite thresholds retained)	+0.1285	+0.3280	0.301 across available attempts
Joint-success, finite-Mondrian-threshold pairs	approximately +0.0275	approximately +0.0013	1.000 within this subset

From budget 50 onward, the finite-threshold sensitivity subset still showed a material coverage–set-size trade-off, and by 250 it approached the primary joint-success conditional result because finite thresholds were nearly complete. Across all budgets, 50/50 task–model–budget cell medians had positive worst-class differences, 38/50 were at least +0.10, and the two model directions agreed in 25/25 task–budget comparisons. Marginal coverage and absolute marginal-coverage error did not have a common improvement direction. The 50 primary cell rows are supplied in `source_data_v2/supp_h5_common50_cells.csv`.

S8. Simulation–natural-shift alignment

The probability simulation crossed seven mechanisms, four prevalences, five budgets, and four calibration methods, with 200 repetitions in each cell. Equal averaging of the frozen isotonic summary cells gave mean log-loss benefits −0.4102, −0.1721, +0.0583, +0.1973, and +0.2397 at budgets 25, 50, 100, 250, and 500. The sign change contrasts with the primary-set natural-shift aggregate, which remained negative at 500. The mechanism grid is therefore simpler than the natural shifts in a scientifically consequential way.

Supplementary Table S7. Isotonic probability-simulation mean effect.

Budget	Mean simulated Δ log loss	Alignment with primary-set real aggregate
25	-0.4102	same negative direction
50	-0.1721	same negative direction
100	$+0.0583$	disagrees in sign
250	$+0.1973$	disagrees in sign
500	$+0.2397$	disagrees in sign

The conformal simulations support a class-asymmetry mechanism and the possibility that classwise coverage requires larger sets. Their stored `finite_threshold_rate` is conditional on estimable successes; the corresponding all-attempt rates were approximately 0.236, 0.424, 0.516, 0.728, and 0.763 over budgets 25–500. They therefore cannot be directly differenced from the natural-shift available-attempt H5 rates. Some reference or overconfidence scenarios raised set cost without material worst-class benefit. Simulation supplies mechanism context only and is not a substitute for an additional natural task.

Supplementary Table S8. Public data provenance, version, and redistribution boundary.

Task family	Public source/version	Entity and outcome audit key	Access/redistribution boundary
ACS PUMS	2018 1-year person and housing CSV files	person or `RELP=0` household reference row; `PINCP`/`FS`	public Census microdata; cite source and do not imply Census endorsement
BRFSS	2015, 2017, 2019, 2021 annual files	respondent; valid diabetes and `_PRACE1` codes only	public CDC files; follow CDC citation and use terms
NHANES	1999–2000 through 2017–2018 public-use cycles	cycle-qualified `SEQN`; `LBXBPB`	public-use files under the NCHS data-use agreement (National Center for Health Statistics 2024)
UCI diabetes	DOI-bound dataset version, accessed for the frozen build	stable-hash patient index encounter; `readmitted != NO`	CC BY 4.0; attribution required (Clore et al. 2014)
College Scorecard	`MERGED2018_19_PP.csv` from the frozen raw-data archive	`UNITID`; `C150_4`	public Department of Education data; release metadata and attribution retained

S9. Reproducibility and source-data map

Formal computation used Python 3.11.15, NumPy 2.4.6, pandas 3.0.3, SciPy 1.17.1, scikit-learn 1.9.0, XGBoost 3.2.0, matplotlib 3.11.0, seaborn 0.13.2, PyArrow 24.0.0, PyYAML 6.0.3, psutil 7.2.2, and Folktables 0.0.12. The server ran CentOS Linux 8 with kernel 4.18.0-348.el8.x86_64, two Intel Xeon Silver 4210R processors (20 physical and 40 logical cores) and approximately 125 GiB RAM. Formal XGBoost execution was CPU-only with `n_jobs=4`. The exact package lock is `environment.lock.yml`; a user-space reconstruction command is `micromamba create --prefix ~/.local/envs/reliable-shift-stage1 --file environment.lock.yml`. The supplied exact-build lock is for `linux-64`. Document conversion used Pandoc 3.10 and Tectonic 0.16.9; PDF inspection used Poppler 26.05.0. These presentation-only tools are recorded separately in `DOCUMENT_TOOLCHAIN.lock.md` and did not fit models or alter experiment outputs.

The base-model and update details needed to re-execute the prespecified model configurations from prepared inputs are fully specified in the main Methods. In particular, logistic-regression candidates used $C = 0.1$ and 1.0; XGBoost candidates used `(min_child_weight, reg_lambda)` equal to `(1,1.0)` and `(5,2.0)`; probabilities were clipped at 10^{-6} only for numerical stability; intercept offsets used Brent roots on `[-40,40]`; and sigmoid/isotonic solver settings are reported verbatim. Split seeds were 0–2, logistic model seed was 0, XGBoost model seeds were 0–1, and adaptation seeds were 0–99.

Each main figure has both PDF and 400-dpi PNG output. Every figure and main table has a CSV source file, and `source_data_v2/SOURCE_DATA_INDEX.csv` records its SHA-256 and transformation. Figure 1 is a protocol diagram; Figure 2 uses the 20 equal-model H2 method–budget summaries; Figure 3 contains the 200 unpooled primary-set task–model–method–budget H3 cells; Figure 4 uses the five archived H5 budget records; and Figure 5 contains the 54 primary H4 pairs. Main Tables 1–4 have independent CSV, Markdown, and LaTeX forms.

The public release contains manuscript files, figures, tables, figure- and table-level source data, analysis and artifact-generation code, environment specifications, checksums, and a documented workflow for one representative end-to-end ACS model shard. An optional workflow re-executes the 54 prespecified model configurations from prepared, audited Parquet inputs. The package does not redistribute the original third-party datasets, reconstruct five of the six task families from raw official downloads, or contain the complete intermediate results tree and downstream analysis pipeline needed to regenerate all article source-data files from that re-execution alone. Data-acquisition instructions identify the official sources and applicable provider terms.

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